Module 4 — Stretch: Shared Chart Gallery & Comparative Analysis

Honors Track. This stretch assignment is not required for program completion but counts toward Honors distinction. Only work on this if you have completed all core assignments for this module and are On Track or Advanced. If you are behind on core work, focus there first.

The Challenge

This stretch has two parts: an **individual gallery contribution** and a **paired comparative analysis**.

Part 1 — Gallery Contribution (individual). Pick your best chart from the lab or integration — or create a new one — refine it to publication quality, and submit it with the code that generates it and a brief write-up explaining the finding.

Part 2 — Comparative Analysis (pairs). Partner with one classmate. Each of you independently picks a different Key Performance Indicator (KPI) from your integration work and creates a polished visualization for it. Then compare: why did you pick different KPIs? Why did you choose different chart types? What do your findings say when read side by side? Write a joint analysis explaining where you agreed, where you disagreed, and what you learned from the comparison.

Then review two other learners' gallery contributions.

What You'll Learn

- Refining a chart from "good enough" to publication quality
- Writing concise data narratives (100 words explaining a finding)
- Defending your analytical choices to a peer and evaluating theirs
- Discovering how two people can look at the same data and reach different conclusions
- Fork-based Pull Request (PR) collaboration workflow (reinforcement from Module 1)

How This Assignment Is Graded

Your TA scores this assignment on 4 dimensions: Chart Quality, Write-up, Comparative Analysis, and Peer Reviews. The comparative analysis carries the most weight — it is the part that requires genuine analytical judgment.

AI screening: Your TA screens submissions for generic, AI-generated content. Submissions flagged as AI-generated receive a **0 with resubmission required**. The comparative analysis is inherently hard to fake — it requires referencing a real conversation with your partner and defending specific choices you made. Write in your own voice.

Setup

1. **Fork** the chart gallery repo: LevelUp-Applied-AI/m4-s2
2. **Clone** your fork locally
3. Create a new branch: `chart-gallery/your-github-username`

Part 1 — Gallery Contribution (Individual)

1. Create your contribution

Create a folder `contributions/your-github-username/` in your fork containing:

File	Content
`chart.png`	Your polished visualization (PNG, at least 800x600)
`generate_chart.py`	The Python script that produces `chart.png` (must run independently)
`README.md`	A 100-word write-up: what data, what finding, why it matters

Chart quality checklist:

- [ ] Finding-based title (states the insight, not just the data)
- [ ] Labeled axes with units
- [ ] Colorblind-safe palette
- [ ] No chart junk (unnecessary gridlines, 3D effects, decorative elements)
- [ ] Annotations on key data points if applicable
- [ ] `fig.savefig('chart.png', dpi=150, bbox_inches='tight')`

2. Open a PR

Push your branch and open a PR to the upstream gallery repo's `main` branch. Your PR description should include:

- The chart image (drag into the PR description on GitHub to embed it)
- Your 100-word finding summary

Part 2 — Comparative Analysis (Pairs)

3. Find a partner

Partner with one other learner working on this stretch. If you need to find a partner remotely, post in the course Slack channel. You will each contribute independently, then write a joint analysis.

4. Independent KPI selection

Each partner independently:

- Picks a **different KPI** from their integration work (do not coordinate which KPI — pick before discussing)
- Creates a polished visualization for that KPI (this can be the same chart as your gallery contribution, or a different one)

5. Compare and analyze

Meet with your partner (in person, Slack, or video call) and discuss:

- **Why did you pick different KPIs?** What made yours feel important?
- **Why did you choose different chart types?** (If you both chose the same type, discuss whether a different type would tell a different story)
- **What do your findings say when read together?** Do they reinforce each other, contradict, or reveal something neither chart shows alone?
- **What would you change after seeing your partner's work?**

6. Write the joint analysis

Write the analysis together and save it as `contributions/your-github-username/COMPARATIVE_ANALYSIS.md` — each partner places a copy in their own contributions folder. Include:

1. **Your KPI and chart** — which KPI you picked and why (2–3 sentences)
2. **Your partner's KPI and chart** — which KPI they picked and why (2–3 sentences)
3. **Where you agreed** — what both analyses confirm about the Amman Digital Market (3–4 sentences)
4. **Where you disagreed** — different conclusions, different priorities, different chart design choices. Explain *why* you see it differently, not just *what* differs (3–4 sentences)
5. **What you learned** — one thing you would change about your own integration analysis after this conversation (2–3 sentences)

Length: 300–500 words total. Both partners submit the same document in their own PR.

Part 3 — Peer Reviews (Individual)

7. Review two peers

Find two open PRs from other learners (not your partner). For each, leave a review with:

- One thing the chart does well
- One specific improvement suggestion (reference the chart type framework, Gestalt principles, or annotation practices from the reading)

Submit

Your single PR should contain everything your TA needs to grade all 4 dimensions:

File in your PR	What it covers
`contributions/your-username/chart.png`	Chart Quality dimension
`contributions/your-username/generate_chart.py`	Chart Quality dimension
`contributions/your-username/README.md`	Write-up dimension
`contributions/your-username/COMPARATIVE_ANALYSIS.md`	Comparative Analysis dimension
2 reviews on other learners' PRs	Peer Reviews dimension (your TA checks the gallery repo)

Your PR description should include:

- The chart image (drag into the PR description to embed it)
- Your partner's GitHub username (so your TA can verify both partners submitted)

Paste your PR URL into TalentLMS → Module 4 → Thursday Stretch. Resubmissions are accepted through Saturday of the assignment week.

Tips

- The best gallery contributions tell a story in one image. A reader should understand the finding without reading the code.
- When reviewing peers, be specific. "Nice chart" is not a review. "The sorted horizontal bars make the comparison easy to read — consider adding value labels at the end of each bar so the exact numbers are visible without hovering" is a review.
- Your `generate_chart.py` should run independently — include any data loading or inline data it needs. Another learner should be able to run it and reproduce your chart.
- The comparative analysis is the most valuable part of this stretch. The point is not to agree with your partner — it is to articulate *why* you made different choices and what that reveals about the data.